

ReportXpert

Autonomous Institutional Intelligence, Accreditation & Regulatory Reporting Engine

1 TITLE AND BRIEF DESCRIPTION OF THE IDEA / MODEL

EXECUTIVE OVERVIEW

Project Title: ReportXpert: Autonomous Institutional Intelligence, Accreditation & Reporting Engine

Core Concept: An on-premise sovereign AI engine powered by a local **NVIDIA DGX supercomputer node** that autonomously compiles, audits, and formats university accreditation dossiers (NAAC, NIRF, MDRF) with 100% data privacy and zero cloud egress.

Every semester, higher education institutions face an administrative bottleneck: compiling hundreds of complex institutional compliance dossiers for statutory accreditation bodies (NAAC, NIRF, MDRF, NBA) and internal appraisal committees. This manual process typically pulls dozens of senior faculty members away from teaching and research for months to search for paper certificates, reconcile spreadsheet formulas, and verify publication citations.

ReportXpert resolves this crisis by establishing a private, personalized Large Language Model (LLM) dedicated exclusively to the university's internal data ecosystem. The system functions as an always-on digital administrative specialist: it autonomously ingests unstructured faculty CVs, published manuscripts, patent certificates, grant award letters, student records, and placement data. It cross-verifies citations against recognized indexing databases (Scopus, Web of Science), computes statutory ratios, audits mathematical accuracy, and generates submission-ready official compliance dossiers in **under 5 minutes**.

Because the entire system runs locally on campus-owned NVIDIA DGX hardware without internet dependency, sensitive student marks, personnel salaries, and proprietary patent drafts remain 100% confidential within university premises.

2 KEY INNOVATIVE FEATURES AND UNIQUENESS

CORE INNOVATIONS



100% Sovereign Privacy Architecture

Zero Cloud Egress

Unlike commercial cloud AI tools (e.g., ChatGPT) that upload institutional records to third-party public servers, ReportXpert operates entirely inside the university's physical data center. Sensitive student grades, faculty salaries, and patent disclosures never leave the campus network (**0.00 KB cloud egress**).



Automated Faculty Research Profiler

CV to Proof

Ingests raw faculty resumes (Word/PDF) and web profiles. The engine automatically extracts research publications, validates DOI indexing against Scopus, maps sanctioned government grants (DST, DBT, CSIR), catalogs awarded patents, and computes Academic Performance Indicator (API) appraisal scores.



Dual-AI Engine (Assistant + Auditor)

Cooperative AI

Employs a cooperative multi-agent architecture: an *Intelligent Coordinator* (Llama 3.2) that understands everyday plain-English requests from professors, paired with an independent *Strict Mathematical Auditor* (DeepSeek-R1) that cross-validates formulas, audits ratios, and detects anomalies before report generation.



Dynamic Multi-Format Template Engine

Word & Excel

Directly generates ready-to-submit official Microsoft Word (`.docx`) books pre-formatted with institutional headers, qualitative narratives (QIM), and tables of contents, as well as multi-sheet Microsoft Excel (`.xlsx`) workbooks matching exact government portal column structures with zero formula errors.

INSTITUTIONAL UNIQUENESS HIGHLIGHTS

VERIFIED ON-PREMISE AIR-GAP

On-Campus DGX Compute

Executes locally on university-owned hardware; operates completely offline with zero data leakage.

Auditable Digital Proof Links

Every extracted statistic is tied directly to clickable digital proof files for inspection teams.

Natural Language Interface

Administrators query in plain English: "Prepare NAAC Criterion 3.4 research dossier" for instant books.

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PROBLEM ADDRESSED / PRACTICAL APPLICATION

REAL-WORLD UTILITY

The Academic Gridlock: During every accreditation and ranking cycle (NAAC, NIRF, MDRF), institutional productivity plummets. 50+ senior professors and staff are diverted from teaching and scholarly research for 3 to 4 months to manually collect paper certificates, chase colleagues for publication lists, and re-type data into spreadsheets. Manual consolidation leads to formula mistakes, uncounted research papers, and misplaced proof files—resulting in downgraded accreditation tiers and lost research grants. Furthermore, uploading sensitive institutional data to commercial public AI services creates severe data leakage risks.

ADMINISTRATIVE AREA	TRADITIONAL MANUAL PROCESS (MONTHS OF STRESS)	WITH REPORTXPRT (AUTONOMOUS ENGINE)
Faculty Time & Effort	50+ professors working late nights for 3 to 4 months	Completed in under 5 minutes by the on-premise AI
Research & Grant Tracking	Missing papers, uncounted citations, lost sanction letters	100% of publications, patents & grants automatically mapped from CVs
Inspection Day Readiness	Panic searching through filing cabinets for misplaced paper proofs	Instant digital audit trail with 1-click links to every original document
Data Security & Privacy	Severe privacy breach risks by using public cloud tools or agencies	100% confidential and secured inside the university's internal server room

Practical Campus Implementation: Internal Quality Assurance Cells (IQAC), Deans, and Department Heads utilize ReportXpert daily as an automated compliance co-pilot. By uploading raw faculty CVs and departmental sheets, university leadership can ask natural-language questions (e.g., *"What is our total Scopus count for 2024?"* or *"Calculate our NIRF student-to-teacher ratio"*) and receive instant, verifiable answers backed by ready-to-submit Word and Excel reports.

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FUTURE POTENTIAL AND SCOPE FOR FURTHER DEVELOPMENT

SCALABILITY & ROADMAP



1. Multi-Campus Federated Intelligence

Architectural scaling to link multiple affiliated colleges and regional satellite campuses under a unified, federated governance model. Central university leadership gains real-time institutional intelligence across all campuses while ensuring each individual institution retains private custody of its raw records.



3. Direct Regulatory API Gateway

Establishing direct, cryptographically authenticated digital submission pipelines to national regulatory bodies (NAAC portal, AISHE, NIRF API), enabling 1-click end-to-end digital compliance as government accreditation bodies adopt automated machine-readable reporting standards.



2. Predictive Accreditation Simulation

Integrating predictive machine learning models that simulate prospective NIRF ranking shifts and NAAC grade impacts before final submission. The system will recommend targeted strategic interventions—such as optimal faculty recruitment targets, specific research citation goals, or student diversity quotas.



4. Multilingual & Regional Language Support

Expanding localized speech-to-text and multilingual NLP to support state universities, rural colleges, and regional vernacular institutions across India, allowing administrative personnel to query and generate institutional documents in Hindi and regional languages.

Hardware: NVIDIA DGX Supercomputer (A100/H100 Tensor Core GPUs)

Models: Llama 3.2 3B (Coordinator) + DeepSeek-R1 7B (Auditor)

Privacy: 100% Local Inference • 0.00 KB Egress